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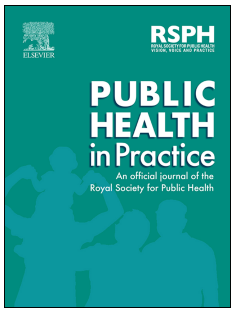

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Correlates of COVID-19 Vaccine Uptake in Ghana: A Case Study

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Declaration of Interest Statement

The authors declare no competing interests.

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abstract

Objectives: COVID-19 pandemic significantly impacted public health and disrupted socioeconomic activities globally. Although vaccination efforts have intensified, overall community uptake of COVID-19 vaccines remain suboptimal. This study explored the facilitators and barriers of vaccine uptake in the New Juaben South Municipality (NJSM) in Ghana.

Study Design: The Exploratory Case Study Design was employed for the research.

Methods: Using concurrent mixed-method approach, quantitative data were collected from 418 randomly selected respondents from eight communities. The qualitative data consisted of responses from 23 participants. Quantitative data were analyzed with SPSS version 28, descriptive statistics, regression analysis, and p-values were used to interpret the findings, whilst participant interviews were recorded, transcribed, and analyzed thematically.

Results: The level of COVID-19 vaccine uptake was 53.1%. Various factors significantly influenced uptake, including age (p=0.053), occupational category (p=0.001), education (p=0.003), income (p=0.001), awareness of effectiveness of vaccines (p=0.002), awareness of ongoing vaccination (p=0.011), infection with coronavirus (p=0.005), close family member infected with coronavirus (p=0.039), death of someone due to COVID-19 (p=0.022) and overall perception towards COVID-19 disease and vaccine (p=0.001). Additionally, respondents above 47 years [AOR=4.99, CI= (1.32-18.91), p=0.018)], health workers [AOR=2.65, CI = (1.03-6.18), p=0.043)], those with basic education [AOR=0.50, CI = (0.27-0.92), p=0.025)], and had good perception towards COVID-19 disease and vaccination [AOR=2.30, CI = (1.37-3.85), p=0.002)], all had higher odds of COVID-19 vaccine uptake. However, deeply rooted religious and cultural beliefs were major barriers to vaccination uptake.

Conclusion: Community-based health communication strategies are essential to counter vaccine misconceptions during pandemics and foster enduring behavioural change.

Keywords:
COVID-19 Vaccine uptake, COM-B model, Vaccine hesitancy, sociocultural barriers, Ghana

1. Introduction

Africa recorded over nine million cases of COVID-19 and almost 176,000 deaths (1). In Ghana, over 171,735 cases and 1,462 cumulative deaths were reported since March 2020 (1). Due to the high transmissibility of COVID-19, vaccines were rapidly developed and distributed globally to counter it (2–5). Despite this, by October 2023, only 36% of Ghana's total eligible population was fully vaccinated; well below the projected 70% target (1). Vaccine hesitancy was among the main causes of the low COVID-19 vaccination coverage in Ghana (6). Studies have shown that many Ghanaians were unprepared to accept the COVID-19 vaccines (7).

Vaccine hesitancy arises from many causes, particularly mistruct in heath systems (Confidence), barriers to access (Convenience), and low risk perception of diseases (Complacency) (8–10). A cross-sectional study by Malesza and Wittman (11), suggested a significant relationship between general health condition, chronic conditions, perceptions of infection, long-term effects, perceived efficacy of vaccines, side effects of vaccines, and attitudes towards vaccination. Similarly, reports from previous vaccine rollouts in Africa show that fear of side effects was largely responsible for low vaccination coverage in the region (12–15).

To fully appreciate the systemic and behavioural predictors of COVID-19 vaccine hesitancy in Ghana, the modified COM-B model of health behavior was adopted to guide this research. COM-B model was developed by Michie et al (16) and is widely used to evaluate vaccine programmes, including the "WHO Tailoring Immunization Programmes Approach" (17). The model conceptualizes vaccine related behaviour as a function of three constructs, namely capability (knowledge, skills, and abilities of individuals), opportunity (external influences, including vaccine manufacturing, infrastructure, pricing and accessibility), and motivation (personal experiences, perceptions of vaccine safety, effectiveness and risks) (16,18,19). We operationalized these variables using unidirectional links for easy comprehension.

2. Methods

2.1 Study setting

The study was conducted in New Juaben South Municipality (NJSM), Eastern Region, Ghana. NJSM had 125,256 residents across eight administrative zones, with two public hospitals, fifteen private hospitals, 34 Community-Based Health Planning S Compounds, four health centers, one polyclinic and one maternity home.

2.2 Study design and Population

We employed the exploratory case study design, with sequential mixed method approach. Quantitative surveys were used to measure the predictors of COVID-19 vaccination, and in-depth interviews used to explore underlying motivations, barriers, and contextual influences. The results were integrated to identify areas of divergence and complementarity (20) (Creswell et al., 2003).

2.3 Sample size, Sampling techniques, and Instruments

The target population includes adult residents of NJSM. A sample size of 424 was calculated using Cochran's formular (1977) (95% CI, 10% nonresponse) from eight randomly selected communities (Old Estate, Social Welfare, Two Streams, Zongo Community C, Residential Area, School Town, Nsukwao Abotanso and Klu Town-from) from NJSM

A structured questionnaire, adapted from previous literature (21,22) and covering respondents' sociodemographic characteristics, vaccine uptake, knowledge, awareness, experiences, and perceptions of COVID-19 was used. Trained researchers assisted respondents. Each respondent took about 10 minutes to complete the questionnaire. Overall, 418 residents completed the questionnaire (98.6%).

Further, 22 participants were purposively sampled from the questionnaire completers and interviewed. The in-depth interviews were conducted using interview guides designed along the research objectives and the COM-B model. Each interview lasting approximately 45 minutes, in the English or Twi languages, and progressed until data saturation was achieved.

2.4 Data processing and analysis

The quantitative data were cleaned, coded, and analyzed using SPSS version 28. Bivariate associations (Chi-square and Fischer's exact tests) aided variable identification for inclusion in a multivariate logistic regression. The results were expressed as adjusted odds ratios (AORs) at $\alpha=0.05$.

The interviews were audio-recorded and transcribed. Transcripts and fieldnotes were translated and analyzed thematically following Braun and Clarke's (23) framework. First, we reviewed the transcripts to identify patterns; searched for deductive and inductive codes and linked them to the data; refined codes into themes for clarity and consistency; and organized them to develop connectors to write the research report.

2.5 Ethical considerations

Ethical approval was obtained from the Ethics Committee for the Humanities (ECH), University of Ghana, with approval number **ECH 182/22-23**. Administrative permission was obtained from the Municipal Health Directorate. Written and verbal consent was taken from all participants. Participation was voluntary.

3. Results

3.1 Statistical analysis

3.1.1 Sociodemographic characteristics of respondents

Table 1 presents the sociodemographic characteristics of respondents. Out of 418 responses, approximately 53% were females. The mean age of respondents was 40±12 years. The mean monthly income was GHS 1182.00 ± 1021.00.

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Table 1: Sociodemographic Characteristics of Respondents

Variable	Frequency	Percentage (%)
Sex		
Female	264	63.2
Male	154	36.8
Age (years)		
40±12 (Mean)		
18-23	16	3.8
24-29	50	12
30-35	134	32.1
36-41	55	13.2
42-47	67	16
>48	96	23
Marital Status		
Single	160	38.3
Married	233	55.7
Divorced	11	2.6
Widowed	14	3.3
Place of Residence		
Rural	6	1.4
Peri urban	86	20.6
Urban	326	78
Occupational category		
Non-Health Worker	371	88.8
Health Worker	47	11.2
Level of Education		
No Formal Education	3	0.7
Basic Education	103	24.6
Secondary Education	138	33
Tertiary Education	174	41.6
Ethnicity		
Akan	208	49.8
Ewe	52	12.4
Krobo	40	9.6
Hausa	31	7.4
Ga-Adangbe	29	6.9
Other	58	13.9
Religion		
Christian	361	86.4
Muslim	57	13.6
Average Monthly Income (GHC)		
1182.00± 1021(Mean)		
<1000	221	52.9
1000-2000	129	30.9
2001-3000	46	11
>3000	22	5.3
Total	**418**	**100**

Source: Field data, (2023)

3.1.2 COVID-19 vaccination uptake

Overall, 53.1% (222/418) received at least one dose of COVID-19 vaccines. The majority (46.8%, n=104/222) had two doses, while another 29.3% (65/222) took booster shots (three or more). Among the unvaccinated, 68.4% (134/196) showed low intention to vaccinate whereas 31.6% (62/196) stated their intention to vaccinate.

3.1.3 Knowledge and observance of COVID-19 preventive protocols

Most of the respondents (98.1%; 410/418) were aware of COVID-19 preventive measures, including hand washing (98.1%; 410/418), coughing or sneezing into arms (98.3%; 411/418), handshaking (98.6%; 412/418), social distancing (98.8%; 413/418), touching face (98.6%; 412/418), staying home (98.6%; 412/418) and wearing face masks (98.6%; 412/418).

3.1.4 Awareness of COVID-19 Vaccination

Nearly all respondents- 96.7% (404/418) were aware of Ghana's COVID-19 vaccination programme. The predominant vaccine side effects reported include fever 77.2% (312/404), headache 74.0% (299/404), muscle pain 68.8% (278/404), pain at the site of injection 59.2% (239/404) and muscle weakness 52.7% (213/404). About 26.5% (107/404) males and 10.6% (43/404) females reported impacts on sexual performance. Also, chest pains 7.2% (29/404), speech loss 7.4% (30/404) and diarrhea 9.9% (40/404) were reported.

3.1.5 Experiences with COVID-19 infection

At least 63% (63/418) were infected but recovered fully. About 84.9% (355/418) were not infected with COVID-19. About 27.6% (98/355) of those not infected indicated that close family members got the disease. Relatives hospitalized due to COVID-19 and those who knew someone who had died from the virus constituted 34.7% (123/355) and 30.3% (108/355), respectively. Finally, about 30.9% (129/418) experienced work or wage disruptions during the pandemic.

3.1.6 Perceptions towards COVID-19 vaccination and infection

The majority 76.3% (319/418) felt the vaccines were safe. About 62.4% (262/418) of doubters felt the vaccine infects people with COVID-19; 76.1% (318/418) believed that COVID-19 virus could still infect people who recovered from the disease. Meanwhile, 67.5% (282/418) believed there was no need for vaccination once a person recovers from the disease. About 75% (313/418) indicated that there was no need for COVID-19 vaccines once they strictly adhered to the pandemic protocols. Overall, 70.6% had adequate knowledge about the vaccines.

3.1.7 Simple logistic regression analysis

Table 2 highlights the results of the simple logistic regression analysis. Significant associations were found between vaccine acceptance and; age (p=0.053), occupational category (p<0.001), Level of education (p=<0.003), monthly income (p<0.001), awareness of effectiveness of vaccine (p=0.002), awareness of ongoing COVID-19 vaccination (p=0.011), infection with coronavirus (p=0.005), close family member infected with coronavirus (p=0.039), death of someone due to COVID-19 (p=0.022) and overall perception towards COVID-19 disease and

vaccine (p<0.001), and uptake of the vaccine. After adjusting for confounding variables, the analysis showed that age, occupational category, level of education, and perception towards COVID-19 disease and vaccine were significantly associated with COVID-19 vaccine uptake. Respondents aged above 47 years were 4.99 times more likely to vaccinate than those aged 18-23 [AOR=4.99; p=0.018]. The analysis also showed that the odds of COVID-19 vaccine uptake among health workers was 2.65 times higher than non-health workers [AOR=2.65; p=0.043)]. Respondents with basic education qualifications were 50% less likely to take up COVID-19 vaccination compared to those with no formal education [AOR=0.50; p=0.025)]. Also, respondents who have a good regard towards COVID-19 vaccines were 2.3 times more likely to take COVID-19 vaccines compared with those who have poor perceptions towards COVID-19 and vaccines [AOR=2.30; p=0.002)].

Table 2: Logistic regression of Factors Influencing COVID-19 Vaccine Uptake in the NJSM

Variable	OR (95% CI)	*p*-value	AOR (95% CI)	*p*-value
Age (years)		**0.053**		
18-23	Referent		Referent	
24-29	3.52 (1.00-12.43)		1.66 (0.42-6.60)	0.472
30-35	3.18 (1.00-10.38)		1.79 (0.48- 6.59)	0.383
36-41	2.50 (0.72- 8.72)		1.10 (0.28- 4.35)	0.896
42-47	3.28 (0.96-11.21)		1.66 (0.43-6.44)	0.462
>47	5.47 (1.64-18.28)		4.99 (1.32-18.91)	**0.018***
Occupational category:		**< 0.001***		
Non-Health Worker	Referent		Referent	
Health Worker	5.93(2.59-13.59)		2.65 (1.03-6.81)	**0.043***
Level of Education:		**0.003***		
No Formal Education	Reference		Referent	
Basic Education	0.52 (0.32-0.86)		0.50 (0.27-0.92)	**0.025***
Secondary Education	0.49 (0.31-0.77)		0.70 (0.41-1.21)	0.204
Tertiary Education	1		1	
Average Monthly Income:		**< 0.001***		
<1000	Referent		Referent	
1000-2000	1.16 (0.75-1.80)		0.90 (0.54-1.51)	0.700
2001-3000	3.24 (1.60-6.60)		2.26 (0.97-5.27)	0.059
>3000	11.46 (2.61-50.20)		5.27 (0.95-29.22)	0.057
Awareness of effective vaccine for COVID-19:		**0.002***		
Not aware	Referent		Referent	
Aware	3.88(1.62-9.30)		1.18 (0.38-3.61)	0.774
Awareness of ongoing COVID-19 vaccination:		**0.011***		
Not aware	Referent		Referent	
Aware	7.17(1.59-32.46)		4.77 (0.76-29.81)	0.094
Infection with coronavirus:		**0.005***		
Never infected	Referent		Referent	
Ever infected	2.30(1.29-4.10)		2.00 (0.90-4.43)	0.088
Close family member ever infected with coronavirus:		**0.039***		

Never infected	Referent	Referent	
Ever infected	1.63(1.02-2.59)	0.74 (0.37-1.50)	0.404
Death of someone you know from COVID-19:	**0.022***		
No death from COVID-19	Referent	Referent	
Death from COVID-19	1.65 (1.08-2.52)	1.18 (0.65-2.12)	0.588
Overall Perception towards COVID-19 Disease and Vaccine:	**< 0.001***		
Poor Perception	Referent	Referent	
Good Perception	3.18 (2.04-4.94)	2.30 (1.37-3.85)	**0.002***

*Statistical significance, p<0.05, OR=Crude Odds Ratio, AOR=Adjusted Odds Ratio; Source: Field data, (2023)

3.2 Findings from the qualitative study

Nine (9) sub-themes emerged from the interviews (see Table 3).

Table 3 COM-B CONSTRUCT

Themes	Subthemes
Capability (Knowledge and skills)	Personal decision and choice based on perceived benefits
	Misinformation and misconceptions about COVID-19 vaccines
	Future concerns and uncertainties around Covid-19 vaccines
	Culture and religious beliefs on COVID-19 vaccine
Opportunity (Physical & Social)	Influence of society and leaders
	Workplace/official requirement
Motivation (Reflective and Automatic)	Public health recommendations
	Protection of loved ones
	Motivation from people who had taken the vaccine

3.2.1 Capability

The findings show that vaccine acceptance was significantly influenced by future concerns and uncertainties, culture and religious beliefs, and misinformation and disinformation, among others.

(i) Personal decision and choice based on perceived benefits: It emerged that individuals' beliefs and circumstances strongly influenced their vaccination decision.

> *Nobody influenced me. I thought it was wise that I must protect myself. So, I took the vaccine to protect myself* (P22)

***(ii) Exposure to information about COVID-19 vaccine*:** Participants were convinced by public health officials about vaccine safety and side effects.

> *I spoke to a couple of medical workers who assured me that the vaccine was safe. They told me about the implications of having sickle-cell and having to get infected with COVID-19. I also listened to the broadcast messages of other public health specialists on television.* (P15)

(iii) Future concerns and uncertainties: Participants questioned the long-term effects of vaccines on humans since the technology used to develop them were unknown, and the vaccines were untested on humans.

> *…we don't know what it will do to us in the future. Its consequences on our health are yet to be tested* (P20)

> *I wouldn't be 100% sure of the safety, because I know coming up with a vaccine goes through a lot of processes. But looking at the time that the COVID-19 emerged and the time that the vaccine was produced, it was kind of very short.* (P13)

> *We didn't know how it would turn out. It was a new thing. It's not like childhood disease vaccine that we know about.* (P18)

Additionally, some participants doubted the effectiveness of the vaccine because some people took it but still reported to health facilities with COVID-19.

> *Some people have not taken the vaccine, yet they are healthy and going about their normal lives. There are others who have taken it and are still infected. So, for me, the COVID-19 vaccine is not effective.* (P8)

(iv) Culture and customary Beliefs: Participants mentioned that longstanding cultural and customary practices were significant considerations in their decision to vaccinate.

> *As for me and my family, we have been taking a mixture of tree barks and some herbs since the pandemic started, and we never got infected, though we did not take the vaccine. Can't you see for yourself that I am strong? Our forefathers used complementary and alternative medicines/traditional herbs to treat all conditions.* (P13)

(v) Misinformation and Misconceptions about the COVID-19 Vaccine*:* Participants were highly influenced by information from the public, particularly from social media.

> *Well, when you watch the WhatsApp videos, TikTok, or Facebook, they often say if you take that injection, you can't give birth or in the long term, women who are menopausal you will start having menses again… and people will develop some diseases later in life.* (P9)

> *…the first story was that the whites are just being wicked. They are introducing the vaccine into our system just to kill Black people so that they can have the power or the authority to come and rule over us.* (P15)

3.2.2 Opportunity

On the physical and social determinants, most of the participants mentioned that requirements from employers, the need to travel, and access to public spaces were major considerations.

(i) Workplace/Official requirements: Several participants expressed the view that workplace or official requirements strongly motivated their decision to vaccinate. Especially in areas where proof of COVID-19 vaccination was required to gain access to public places or emigrate.

> *Well, they came up with this regulation that if you don't take the vaccine, you can't travel. There are some social gatherings and some places you cannot go. For instance, at the airport, even when you are going to escort someone, if you don't have the COVID-19 vaccination card with you, you have no right to hang around the airport*. (P9)

> *For me, at my workplace, it was compulsory for everyone to receive the vaccine. Not taking the vaccine was like going against company policy. .* (P15)

(ii) Influence of society and leaders: Participants discussed the role of influential members of society such as politicians, community leaders, business leaders and celebrities on ordinary citizens. Leaders' public education boosted vaccine support.

> *At a point, my presiding church elder, will invite the people doing the vaccination to the church and make an announcement for people to get vaccinated.* (P10)

> *Some politicians and businessmen donated several boxes of nose masks and encouraged people to go and vaccinate to stay safe. We saw our president, our chiefs, and other influential people receiving it on live camera and that motivated a lot of us.* (P11)

3.2.3 Motivation

Participants expressed both emotive and reflective drivers. The key highlights include the need to protect loved ones from infection.

(i) Motivation from People who have taken the vaccine: Participants described how experiences and recommendations by people who had received the vaccine influenced them.

> *The number one reason why I decided to take the vaccine was because some people had it and they shared their experiences of not suffering any adverse effect. (P2*)

***(ii) Protection of significant others*:** The desire to protect relatives and loved ones from infection was another driving force behind vaccine uptake.

> *As the breadwinner and the one who was always going out for food and other essentials at the peak of the pandemic and lockdown, I needed to save my life and protect my family*. (P7)

4 Discussion

The study assessed the predictors of COVID-19 vaccination in NJSM. Our analyses found old age, educational level, health worker status and positive COVID-19 perception to be significant uptake predictors (Table 2). This mirrors previous studies in Nigeria and Ghana where lower vaccination rates was recorded among non-health workers (24,25), likely due to lower virus-related knowledge. Surprisingly, and contrary to other studies in LMICs (26), participants with basic education in this research were 50% less likely to vaccinate compared with those with no formal education (AOR=0.50, CI=0.27- 0.92, p=0.025).

Also, although vaccine hesitancy was low (32%) compared to 80.9% in Nigeria (24) and 62.7% in other parts of Ghana (27), uptake to full dosage reached only 53.1%, showing lack of trust. Concerns about side effects and efficacy hinder acceptance (26) and participants reporting fever (76.8%), headache (73.6%), muscle pain (68.5%), pain at the site of injection (58.9%) and muscle weakness (52.5%) as side effects. Similarly, in a study of Ghanaian university students by Darko, et al (28), low perception of severity of infection, fear and doubt were strong predictors of vaccine hesitancy. Rzymski et al. (29) suggested that even with 68% vaccination acceptance, people would be hindered by side effects, low efficacy and testing mistrust (30,31).

Further, vaccine misconceptions and misinformation, facilitated by religious and cultural beliefs was a major barrier (32). For instance, similar to findings of this research, in Malawi, individuals mentioned that church officials preached that immunization was associated with devil worship (33,34). In Uganda, officials propagated misinformation, fueling skepticism about COVID-19 vaccines (33).

Vaccine uptake in NJSM was also influenced by motivation from leaders in society. Political leaders, religious leaders, and traditional leaders had significant influences on participants' decision to vaccinate. This is consistent with findings of Lazarus et al (2021) who also noted that social and frontline workers, as well as trust in governments, promote vaccine acceptability and contribute to public vaccination compliance. In parallel, Haldane and Butler (32) observed that in Malawi and Uganda, immediate relative and friends' vaccine acceptance gives approval for others in the family to vaccinate. In Kenya, those who avoided vaccines were getting their knowledge from social media, community meetings such as barazas (meetings with chiefs), and trusted local and religious leaders. Adedeji-Adenola et al. (24) stated that there were trust

issues over whether the Nigerian government had ensured the safety of the vaccination. A fear that dissuaded many from taking the vaccine although the President of Nigeria advised all Nigerians to comply. To counter this, vaccine accessibility through scheduled campaigns and mobile outreach by healthcare professionals is required.

Vaccine uptake to full dosage was low (53.1%), likely because of the lack of trust and confidence in the jabs (53.1%). Although some participants held positive perceptions of the benefits of the COVID-19 vaccines, others were demotivated by their safety and effectiveness concerns. This finding was confirmed by a study conducted in Poland which similarly revealed varying levels of faith in licensed vaccines throughout Europe (29). Similarly, the findings of this research revealed that residents of NJSM who had positive perceptions of COVID-19 disease and vaccine were 2.30 times more likely to be vaccinated compared to those with negative perceptions (AOR=2.30, CI=1.37–3.85, p=0.002). Those who were unwilling to accept the jabs generally expressed negative opinions on the safety and effectiveness of the vaccine (35,36). Adedeji-Adenola et al. (24) stated that there were trust issues over whether the Nigerian government had ensured the safety of the vaccines, a fear that dissuaded many from taking the jabs against the President's advice for all Nigerians to vaccinate.

Limitations

The case study design limited generalizability of our findings to other parts of Ghana. However, in areas with similar contextual issues, our findings may be applicable. Future studies should have broader scopes, covering multiple regions in the country to examine the complex socio-cultural and economic predictors of vaccine uptake nationally.

5. Conclusion

COVID-19 vaccination uptake in Ghana is linked to older age, health worker status, educational level and having good perception about vaccines. Key facilitators of vaccine uptake included motivations from people who had taken the vaccine, protection of significant others, workplace/official requirements, exposure to COVID-19 vaccine information and the influence of leaders in the society. In addition to efforts to improve physical access to vaccine doses, ccomprehensive community-based public health communication strategy is crucial to counter COVID-19 vaccination misconceptions and to foster behavioural change.

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Data availability statement

Research data is available upon request.

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Declaration of Interest Statement

The authors declare no competing interests.

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☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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